

E-Commerce Sales Dashboard Using Power BI

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Course: Data Analytics / Business Intelligence

1. Introduction

The purpose of this project is to analyze e-commerce sales data using Microsoft Power BI and create an interactive dashboard for business decision-making. The dashboard provides insights into sales performance, customer orders, payment methods, product revenue, referral sources, and order status.

2. Objectives

The main objectives of this project are:

- To import and clean sales data.
- To create key performance indicators (KPIs).
- To analyze revenue and order trends.
- To identify the most profitable products.
- To study customer payment preferences.
- To evaluate referral sources and order status distribution.

3. Dataset Description

The dataset contains e-commerce transaction records with information related to customer orders and sales.

Important Fields

- OrderID
- CustomerID
- Product
- Quantity
- UnitPrice

- TotalPrice
- Date
- PaymentMethod
- OrderStatus
- ReferralSource

The dataset contains approximately 1,200 records.

4. Data Preparation and Cleaning

The dataset was imported into Power BI using the Excel connector. Data cleaning was performed in Power Query Editor.

Cleaning Steps

1. Imported Excel dataset into Power BI.
2. Verified column headers.
3. Checked data types for all columns.
4. Converted Date field into date format.
5. Verified numeric fields such as Quantity, UnitPrice, and TotalPrice.
6. Removed data inconsistencies where necessary.
7. Loaded the cleaned data into Power BI for visualization.

5. Dashboard Development

Several visualizations were created to analyze sales performance.

KPI Cards

Total Orders

Displays the total number of orders processed.

Total Revenue

Shows the overall revenue generated from sales.

Average Order Value

Represents the average amount spent per order.

Revenue by Product

A bar chart was created to compare revenue generated by different products. This visualization helps identify the highest-performing products.

Sales Trend Over Time

A line chart was used to analyze revenue trends across different dates. It provides insights into sales fluctuations and business growth patterns.

Orders by Payment Method

A column chart was created to display the number of orders made through different payment methods such as:

- Online Payment
- Cash
- Credit Card
- Debit Card
- Gift Card

Order Status Distribution

A donut chart was used to visualize order status categories including:

- Delivered
- Shipped
- Pending
- Returned
- Cancelled

Orders by Referral Source

A bar chart was created to analyze customer acquisition channels such as:

- Instagram
- Email
- Google
- Facebook
- Referral

6. Dashboard Insights

Based on the analysis, the following observations were made:

1. Total Orders were approximately 259.
2. Total Revenue generated was approximately 275.29K.
3. Average Order Value was approximately 1.06K.
4. Chair and Printer products generated higher revenue compared to other products.
5. Online payments were among the most frequently used payment methods.
6. Instagram and Email contributed significantly as referral sources.
7. Delivered orders represented the largest portion of order status distribution.
8. Sales showed fluctuations over time, indicating seasonal or demand-based variations.

7. Conclusion

The E-Commerce Sales Dashboard was successfully developed using Microsoft Power BI. The dashboard provides valuable business insights through interactive visualizations and KPI metrics. It enables users to monitor sales performance, customer behavior, payment preferences, and product profitability. Such dashboards help organizations make informed business decisions and improve overall operational performance.